Savvas Paragkamian

Microbial ecology of extreme and subterranean environments: rope-access sampling, genome-resolved analysis, FAIR data
Postdoctoral researcher, IMBBC–HCMR

s.paragkamian@gmail.com · +30 6934008102 · Irakleio, Crete, Greece · Hellenic nationality
ORCID 0000-0002-8508-2521 · github.com/savvas-paragkamian · LinkedIn

Education

2024 **PhD**, "Deciphering the relation of the microbiome interactome with ecosystem function and biogeochemical processes". University of Crete, Department of Biology, and Institute of Marine Biology, Biotechnology and Aquaculture (IMBBC), HCMR. DOI: 10.12681/eadd/57558
Committee: Prof. P. F. Sarris, Dr E. Pafilis, Prof. I. Antoniou

2017 **MSc in Complex Systems and Networks**, Aristotle University of Thessaloniki. Grade 9.58/10 (Excellent). Supervisors: Prof. I. Antoniou, Prof. S. Sgardelis. Thesis: Exploring the centrality–lethality rule in signed protein networks.

2015 **BSc in Biology**, University of Crete. Grade 7.30/10 (Very Good). Supervisor: Prof. C. Nikolaou. Thesis: Menzerath–Altmann's law in human gene families at the gene–exon level.

Awards and Fellowships

EMBL CPP Fellowship 2026 EMBL Corporate Partnership Programme Fellowship for visiting scholars, hosted by the Korbel and Bork groups, EMBL Heidelberg. Integrated multi-omic analysis – metagenomes, metatranscriptomes and metabolomes – of ∼1,200 Traversing European Coastlines (TREC) soil samples, to characterise soil health along European coastlines.

Dawn Field Award 2024 Dawn Field Award for Outstanding Contributions to Genomic Standards, Genomic Standards Consortium, for contributions to the genomics and island microbiome research communities.

PhD Scholarship 2022–2024 Competitively awarded personal research funding: 3rd call for Doctorate Scholarships (no. 5726), Hellenic Foundation for Research and Innovation and General Secretariat for Research and Innovation.

Travel Grant 2024 Qlife Quantitative Biology Winter School, PSL, Paris.

Research Experience

2026 – present **Postdoctoral researcher, KYBELE.** Knowledge Yield from Biodiversity Literature through Large Language Model extraction: applying LLMs to extract structured biodiversity knowledge from the literature at scale. ELIXIR. Host: IMBBC – HCMR. PI: Mr S. Ninidakis.

2024 – 2025 **Postdoctoral researcher, Biosolutions.** Sustainable solutions addressing the effects of climate change on crop tolerance to biotic and abiotic stresses. Using endophytic and Gourgouthakas Cave isolates as plant growth promotion and biocontrol agents against major crop pathogens. Host: University of Crete. PI: Prof. P. F. Sarris.

2024 – 2025 **Data scientist, Invertebrates Monitoring.** Evaluation of the conservation status of protected species and habitats in Greece; Invertebrates group for Art. 17 of Directive 92/43/EEC. Developed the ETL workflow for FAIR and reproducible invertebrate assessments. Host: Natural Environment and Climate Change Agency (N.E.C.C.A.). PI: Dr K. Triantis.

2022 – present **Island Microbiome Encyclopaedia.** Shotgun metagenomics of the Island Sampling Day Crete soil collection; co-led field sampling and data analysis. PI Dr N. Kyrpides (JGI, Award DOI), Genomic Standards Consortium, HCMR.

2021 – 2022 **EMODnet Biology Phase III & IV.** Text mining on marine taxa and traits. EU Maritime and Fisheries Fund. Host: IMBBC. PI: Ms D. Mavraki.

2020 – 2021 **PREGO.** Development of PREGO, a literature and data integration platform associating organisms and environments with processes. Hellenic Foundation for Research and Innovation. Host: IMBBC. PI: Dr E. Pafilis.

2020 – 2021 **CMBR.** Centre for the Study and Sustainable Exploitation of Marine Biological Resources, a large-scale Greek Research Infrastructure of the National Roadmap. NSRF 2014–2020. Host: IMBBC. PI: Dr A. Magoulas.

2017 – 2019 **Conservation of the Cave Fauna of Greece.** Development of the Cave Fauna of Greece database. MAVA Foundation pour la Nature, WWF Greece. Host: INSPEE-HISR. PI: K. Paragamian.

2015 – 2018	**EU Rural Development – LEADER.** Co-organised 3 projects documenting (description, mapping, audiovisual), re-rigging and community engagement for the canyons of Crete. Host: INSPEE-HISR. PIs: K. Paragamian and S. Paragkamian.

Funded Proposals Co-authored

TETTRIX	Transforming European Taxonomy through Training, Research and Innovations. EU Horizon Europe. PI: Dr I. Stathi, NHMC.
KYBELE	Knowledge Yield from Biodiversity Literature through LLM Extraction. ELIXIR. PI: Mr S. Ninidakis, HCMR.
CCMRI	Climate Change Metagenomic Record Index, a dynamic researcher-notification and sample classification system. GSRI–HFRI. PI: Dr E. Pafilis, HCMR.

Teaching, Supervision and Events

2026	Co-supervision, **undergraduate thesis** of Ms P. Kapouralou: characterisation and properties of seven new *Janthinobacterium* strains from Gourgouthakas Cave. University of Crete.
2026	TETTRIS: Soil Data Science, Standards and Integration. Online course, students and general public. EU Horizon.
2025	Workshop organising team: Scientific Dialogue on Specific Conservation Objectives for Natura 2000 areas. NECCA.
2025	Co-supervision, **MSc thesis** of Ms N. Kokolaki: beneficial plant microbes against drought and high-salinity abiotic stress. University of Crete.
2018, 2025, 2026	Network Science Workshop, course BC204, Bioinformatics MSc Programme, University of Crete.
2024	Field-Work Methods in Speleology: different data and standards. Aristotle University of Thessaloniki, Centre for Education and Lifelong Learning.
2024	Workshop organising team: EMBL TREC Crete; Marine Biodiversity and One Health. IMBBC.
2020	GEO-IN: Geotourism in insular geoparks. Workshop for general public and entrepreneurs. Interreg V-A Greece–Cyprus.

Field Expeditions and Sampling

Island soil microbiomes and endophytes	Co-organised Island Sampling Day Crete 2022 (29 participants, 72 sites, 435 samples). JGI – GSC – HCMR. Also co-organised Chryssi island samplings in 2021 for isolating endophytic microbes from halophytes, PI Dr P. Sarris, University of Crete and IMBB-FORTH.
Cave microbiome	Vertical microbiome sampling of Gourgouthakas Cave, the deepest cave of Greece (−1,100 m). University of Crete.
TREC expedition	EMBL Traversing European Coastlines, Athens superstop, 2024. Contributed sampling permits, logistics, soil collection and science communication.
Cave arthropods	Over 100 cave samplings across Greece. INSPEE-HISR. Also cave samplings in archaeological sites for BIAS II funded by the Ministry of Culture and Ministry of Environment.
Raptor birds	More than 50 nestlings captured on cliffs by rope access (vertical ecology): Bonelli's eagle, Golden eagle, *Gypaetus barbatus*, across the Aegean and Crete. Natural History Museum of Crete.

Skills

Languages	Greek (native), English (C2), French (A2).
Research methods	FAIR–by–design sampling for expeditions; amplicon and full-length 16S rRNA profiling; metagenomic microbiome workflows (sequences to numerical ecology); comparative and genome-resolved analysis; biosynthetic gene cluster (BGC) mining; complex network analysis; statistical and mathematical modelling; geospatial analysis (GIS); species hotspots and conservation assessments; data curation and submission; text mining and agentic information retrieval; data integration ETL workflows.
Programming	**Proficient:** R, AWK. **Intermediate:** Python, Bash. **Basic:** Julia.
Tools	Unix, Git, Vim, Tmux, Conda, HPC (Slurm), containers (Podman, Docker), Tidyverse, R Targets, r-spatial, QGIS, LaTeX, local and agentic LLM tooling (Ollama, Claude Code, OpenCode).

Data resources	**Biodiversity:** GBIF, MicrobeAtlas. **Taxonomy:** GBIF, NCBI Taxonomy, GTDB, SILVA. **Sequences:** ENA, GenBank. **Literature:** PubMed, OpenAlex, BHL, Consensus. **Remote sensing:** Copernicus, NASA MODIS. **Other:** JGI IMG/M, European Soil Data Centre.
Standards	DarwinCore, GSC MIxS, GloSIS, INSPIRE, DublinCore, IIIF, POSIX.
Logistics	Expertise in organising sampling in extreme environments, *i.e.* high cliffs, deep caves and canyons with waterfalls; coordinating expeditions of 30+ people and team building; assistant in equipment and production for Heinz Troll Photography (2018–2026).
Coordination	Project administration, documentation, minutes and reporting; science communication (press, public talks); meeting and workshop organisation (programme, materials, technical set-up).

Certifications

Vertical	IRATA Level 1 rope access technician (X/306777); certified canyoning instructor, French Federation of Speleology (15 seminars delivered); caving explorer and instructor.
Marine	SSI Master Rescue Scuba Diver; speedboat licence (Hellenic Coast Guard).
Vehicle	Car and motorcycle driving licences; UAV operator (A1/A2/A3).

Databases, Platforms and Workflows Developed

CFG database	The first complete national open database of cave biodiversity. Developed and curated at INSPEE-HISR.
NECCA Art. 17 pipeline	ETL workflow for FAIR and reproducible conservation status assessments of invertebrates under Directive 92/43/EEC.
PREGO	Linking microbes, environments and processes through data integration and text mining.
DECO	Detect–Extract–Curate–Occurrences: workflow for the rescue of biodiversity data from literature to occurrences based on Darwin Core standards.

Selected Conference Presentations

2026, Romania	**Paragkamian S.** *et al.* Full-length 16S rRNA profiling of Gourgouthakas Cave (−1,100 m), Crete; and **Paragkamian S.**, Paragamian K. Mapping and safeguarding the subterranean biodiversity of Greece. 27th International Conference on Subterranean Biology, Mangalia.
2025, Greece	**Paragkamian S.** *et al.* A thousand meters deep: vertical profiling of the subterranean microbiome of Gourgouthakas Cave. Joint MIKROBIOKOSMOS–CEESME Conference, Thessaloniki.
2024, online	**Paragkamian S.** *et al.* From micro to macro: Crete island biodiversity and its soil microbial interactome. 2nd Young Soil Researchers Forum, EUSO Stakeholders Forum.
2024, Greece	**Paragkamian S.** *et al.* From microbes to ecosystems: towards a Crete island soil system. Soil Health: Current Status and Future Needs, Chania.
2024, Germany	Pafilis E., **Paragkamian S.** TREC superstop at Athens. TREC Symposium, EMBL Heidelberg.
2024, France	**Paragkamian S.** *et al.* Defining Crete's macroecology based on its soil microbial interactome. Qlife Quantitative Biology Winter School, Paris (travel grant).
2023, Greece	**Paragkamian S.** *et al.* From micro to macro, defining Crete's macroecology based on its soil microbial interactome. 10th International Conference of MIKROBIOKOSMOS, Larisa.
2023, Greece	**Paragkamian S.** Integrating data and methods: what can we learn about the soil ecosystems of Crete. EMBO Satellite Workshop on Ecology Informatics, Irakleio.
2021, Greece	**Paragkamian S.** *et al.* PREGO: mining literature and -omics (meta)data to associate microorganisms, biological processes and environment types. 9th International Conference of MIKROBIOKOSMOS, Athens.
Additional	5 further oral presentations and 10 posters at international conferences and workshops.

Science Communication and Press

2026	"An underground world of unique biodiversity in the Aggitis Cave" (APE); "Deep inside a Cretan cave, scientists found a hidden world" (Vima Science); "Safety goes out the window" (Kathimerini).

2025 "Aerial rescue falling short" (Kathimerini); "Microbes in the Underground" (Ypokosmos Caving Journal); "Savvas Paragkamian receives the Dawn Field Award" (Vima Science); soil microbes of Crete interviews (Cretalive; Hxonews radio).

2022 Microbe sampling in Gourgouthakas Cave (Cretalive).

Publications

Author name in **bold**; * denotes joint first authorship. ORCID 0000-0002-8508-2521.

Peer-reviewed journal articles

1. Arapitsas, N. P., Christakis, C. A., **Paragkamian***, S., Soultatos, S., Reden, F., Psarologaki, C., *et al.* Unravelling the genomic and functional arsenal of Bacilli endophytes from plants with different lifestyles and their antimicrobial potential. *Microbial Genomics* **12.** doi:10.1099/mgen.0.001801 (2026).
2. Bolanakis, G., **Paragkamian**, S., Maroulis, L., Kontos, P., Poursanidis, D., Nikolakakis, M., *et al.* Weaving together the past, present and future of Greek arachnofauna: Biodiversity, knowledge gaps and conservation. *Insect Conservation and Diversity* **19,** 158–171. doi:https://doi.org/10.1111/icad.70019 (2026).
3. Loukas, A., Kalaentzis, K., Venetsianou, N. K., Damianou, C., **Paragkamian**, S., Lagani, V., *et al.* CCMRI: a classification and curated database of climate change-related microbiome studies. *Scientific Reports.* doi:10.1038/s41598-026-51914-z (2026).
4. Venetsianou, N. K., **Paragkamian**, S., Kalaentzis, K., Loukas, A., Damianou, C., Lagani, V., *et al.* LLM-Assessed Relatedness of Microbiome Study Descriptions Aligns more Strongly with Functional than with Taxonomic Profile Similarity. *Microbial Ecology.* doi:10.1007/s00248-026-02730-5 (2026).
5. Holm, J. B., **Paragkamian***, S., Humphreys, M., Chatterjee, A., Yarwood, S., Gaimaro, J., *et al.* First island-wide, single-day soil collection study on Crete reveals environmental drivers of microbial diversity. *Environmental Microbiome* **20,** 94. doi:10.1186/s40793-025-00752-z (2025).
6. Bolanakis, G., **Paragkamian***, S., Chatzaki, M., Kotitsa, N., Kardaki, L. & Trichas, A. The conservation status of the Cretan endemic Arthropods under Natura 2000 network. *Biodiversity and Conservation.* doi:10.1007/s10531-024-02877-y (2024).
7. Kokoli, M., Karatzas, E., Baltoumas, F. A., Schneider, R., Pafilis, E., **Paragkamian**, S., *et al.* Arena3Dweb: interactive 3D visualization of multilayered networks supporting multiple directional information channels, clustering analysis and application integration. *NAR Genomics and Bioinformatics* **5.** lqad053. doi:10.1093/nargab/lqad053 (May 2023).
8. **Paragkamian***, S., Sarafidou, G., Mavraki, D., Pavloudi, C., Beja, J., Eliezer, M., *et al.* Automating the Curation Process of Historical Literature on Marine Biodiversity Using Text Mining: The DECO Workflow. *Frontiers in Marine Science* **9.** doi:10.3389/fmars.2022.940844 (2022).
9. Zafeiropoulos, H., **Paragkamian***, S., Ninidakis, S., Pavlopoulos, G. A., Jensen, L. J. & Pafilis, E. PREGO: A Literature and Data-Mining Resource to Associate Microorganisms, Biological Processes, and Environment Types. *Microorganisms* **10.** doi:10.3390/microorganisms10020293 (2022).
10. Baltoumas, F. A., Zafeiropoulou, S., Karatzas, E., **Paragkamian**, S., Thanati, F., Iliopoulos, I., *et al.* OnTheFly2.0: a text-mining web application for automated biomedical entity recognition, document annotation, network and functional enrichment analysis. *NAR Genomics and Bioinformatics* **3.** lqab090. doi:10.1093/nargab/lqab090 (Oct. 2021).
11. Fanini, L., Defeo, O., Elliott, M., **Paragkamian**, S., Pinna, M. & Salvo, V.-S. Coupling beach ecology and macroplastics litter studies: Current trends and the way ahead. *Marine Pollution Bulletin* **173,** 112951. doi:https://doi.org/10.1016/j.marpolbul.2021.112951 (2021).
12. Zafeiropoulos, H., Gioti, A., Ninidakis, S., Potirakis, A., **Paragkamian**, S., Angelova, N., *et al.* 0s and 1s in marine molecular research: a regional HPC perspective. *GigaScience* **10.** giab053. doi:10.1093/gigascience/giab053 (Aug. 2021).

Preprints under review

13. Michalopoulou, V. A., Soultatos, S. K., **Paragkamian**, S., Arapitsas, N. P., Margaritopoulou, T., Triviza, M.-F., *et al.* Halophytic endophytes provide transferable immune functions for crop resilience. *bioRxiv.* doi:10.64898/2026.07.21.739774 (2026).
14. Nicolosi, G., Bellvert, A., Amorim, I. R., Balázs, G., Biró, A., Borges, P. A. V., *et al.* An open occurrence dataset for European subterranean biodiversity. *EcoEvoRxiv.* Preprint (2026).
15. **Paragkamian***, S., Christakis, C. A., Michalopoulou, V. A., Plakogiannaki, M., Soultatos, S., Arapitsas, N. P., *et al.* A Thousand Meters Deep: Vertical Profiling of the Subterranean Microbiome of Gourgouthakas Cave. *bioRxiv.* doi:10.64898/2026.03.19.712943 (2026).

Doctoral thesis and book

16. **Paragkamian**, S. *Deciphering the relation of the microbiome interactome with ecosystem function and biogeochemical processes* PhD thesis (University of Crete, Greece, 2024). doi:10.12681/eadd/57558.
17. **Paragkamian**, S. & Paragamian, K. *Ha Gorge: Exploring a Spectacular Hidden Landscape* 64 pp. ISBN: 978-618-82232-1-9 (Irakleio, Crete, Greece, 2015).